THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 6 — SCIENCE · HARD LEVEL CHAPTER TEST SET 2

Chapter 4: **Exploring Magnets** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. A pin stuck to a magnet can also pick up another pin because: 1

(a) air presses them **(b)** pins have glue
(c) the pin acts as a temporary magnet **(d)** the pin is sticky

2. Which of these is the correct way to hold a compass while taking a reading? 1

(a) in a metal box **(b)** close to a speaker
(c) flat on the palm, away from iron objects **(d)** tilted near a steel desk

3. An iron nail dangling from a magnet picks up another nail. The second nail is held because: 1

(a) air presses
(b) gravity
(c) the first nail became temporarily magnetised
(d) glue is present

4. Which method would give the strongest magnet from an iron nail? 1

(a) stroking in one direction repeatedly **(b)** painting
(c) stroking back and forth **(d)** heating

5. Which technique best shows that magnetic force is non-contact? 1

(a) the pin is round
(b) the pin moves toward a magnet held at a distance
(c) the pin sticks after touching
(d) the magnet is heavy

6. After removing the magnet, the chain of nails falls apart. This shows the nails: 1

(a) were heavy **(b)** were plastic
(c) were permanent magnets **(d)** were temporary magnets

7. Which graph-like description fits the pull of a magnet on a pin as distance increases? 1

(a) pull increases **(b)** pull becomes infinite
(c) pull stays same **(d)** pull decreases rapidly

8. A student hangs two bar magnets side by side with N facing N. They move apart. What does this show? 1

(a) gravity **(b)** no force
(c) repulsion between like poles **(d)** attraction

9. Why is soft iron not used to make a permanent magnet? 1

(a) it is shiny **(b)** it is brittle
(c) it is costly **(d)** it loses magnetism easily

10. Why can we not isolate a single magnetic pole? 1

(a) poles fall off
(b) poles are rare
(c) breaking a magnet always yields two-pole magnets
(d) poles are heavy

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. Why do ships carry more than one compass? 2

12. A plastic comb rubbed on dry hair picks paper but not iron nails, whereas a magnet picks nails. Explain. 2

13. How would you test whether a nail is a permanent magnet or a temporary one? 2

14. Describe how a magnet helps in removing iron splinters from the eye safely. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Explain the behaviour of a magnetic needle when a magnet is placed to the east of it. 3

16. Three coins look the same. One is iron, one copper, one aluminium. How would you identify the iron? 3

17. Why is it unsafe to have powerful magnets near children? 3

OR

Explain how a refrigerator magnet and a magnetic board work. 3

18. Why do some magnets have a coating? 3

OR

Compare the magnetic force and the force of gravity on a pin near a magnet. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. Explain how magnetic induction makes a chain of paper clips hang from a magnet and why the chain collapses when the magnet is removed. 5

OR

Explain how a toy that uses magnets may fail and how to correct it. 5

20. Discuss the importance of magnets in modern life and give examples from home, hospital and industry. 5

OR

Discuss how we know the Earth has a magnetic field using three pieces of evidence. 5

END OF QUESTION PAPER — Total: 40 Marks